OS ORANGE LEVEL-2- REPORT

SRN: PES1UG24AM270

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Section: AIML-E

PHASE 1:

1A.

```
shriya@DESKTOP-6VL9DB0:/mnt/c/Users/Shriya/Desktop/PES1UG24AM270-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

1B.

```
shriya@DESKTOP-6VL9DB0:/mnt/c/Users/Shriya/Desktop/PES1UG24AM270-pes-vcs$ ls .pes/objects/d5/
8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
```

PHASE 2:

2A.

```
shriya@DESKTOP-6VL9DB0:/mnt/c/Users/Shriya/Desktop/PES1UG24AM270-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

2B.

```
shriya@DESKTOP-6VL9DB0:/tmp/tree_demo_clean$ find .pes/objects -type f
.pes/objects/dc/cf003a666d66ba0fb761d43512fc1e21f8c2f081cfbad78732df7f2353fb52
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/cc/fb155377cee2f9ddda1473e9e3ad49975883c3b95f1b9da318222921aed861
shriya@DESKTOP-6VL9DB0:/tmp/tree_demo_clean$ find .pes/objects -type f -exec ls -lh {} \;
-rw-r--r-- 1 shriya shriya 180 Apr 16 17:34 .pes/objects/dc/cf003a666d66ba0fb761d43512fc1e21f8c2f081cfbad78732df7f2353fb52
-rw-r--r-- 1 shriya shriya 15 Apr 16 17:34 .pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
-rw-r--r-- 1 shriya shriya 20 Apr 16 17:34 .pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
-rw-r--r-- 1 shriya shriya 98 Apr 16 17:34 .pes/objects/cc/fb155377cee2f9ddda1473e9e3ad49975883c3b95f1b9da318222921aed861
shriya@DESKTOP-6VL9DB0:/tmp/tree_demo_clean$ TREE_FILE=$(find .pes/objects -type f -exec ls -s {} \; | sort -n | head -1 | awk '{print $2}')
shriya@DESKTOP-6VL9DB0:/tmp/tree_demo_clean$ echo "Tree object: $TREE_FILE"
Tree object: .pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
shriya@DESKTOP-6VL9DB0:/tmp/tree_demo_clean$ xxd $TREE_FILE | head -20
00000000: 626c 6f62 2031 3200 6865 6c6c 6f20 776f  blob 12.hello wo
00000010: 726c 640a                                rld.
```

PHASE 3:

3A.

```
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  .gitignore
  untracked:  commit.c
  untracked:  commit.h
  untracked:  index.c
  untracked:  index.h
  untracked:  Makefile
  untracked:  object.c
  untracked:  pes.c
  untracked:  pes.h
  untracked:  README.md
  untracked:  run_tree
  untracked:  run_tree.c
  untracked:  test_objects.c
  untracked:  test_sequence.sh
  untracked:  test_tree.c
  untracked:  tree.c
  untracked:  tree.h

shriya@DESKTOP-6VL9DB0:/mnt/c/Users/Shriya/Desktop/PES1UG24AM270-pes-vcs$
```

```
shriya@DESKTOP-6VL9DB0:/mnt/c/Users/Shriya/Desktop/PES1UG24AM270-pes-vcs/test_repo$ ../pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)
```

3B.

```
shriya@DESKTOP-6VL9DB0:/mnt/c/Users/Shriya/Desktop/PES1UG24AM270-pes-vcs$ cat .pes/index
100755 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776324893 6 file1.txt
100755 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776324894 6 file2.txt
```

```
shriya@DESKTOP-6VL9DB0:/mnt/c/Users/Shriya/Desktop/PES1UG24AM270-pes-vcs/test_repo$ cat .pes/index
100755 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776341920 6 file1.txt
```

PHASE 4:

4A.

```
shriya@DESKTOP-6VL9DB0:/tmp/pes_test_phase4$ /mnt/c/Users/Shriya/Desktop/PES1UG24AM270-pes-vcs/pes log
commit 576426c0f5bb20d6afcb2243f286b17d9aea96fad25b082e430e21e743efb607
Author: Shriya <PES1UG24AM270>
Date:   1776344772

    Update multiple files

commit 9c44176122e919122b1fbdcef57aff338580c6cb56facf93d2a1602e4455c794
Author: Shriya <PES1UG24AM270>
Date:   1776344754

    Add C source file

commit cf52ffc93e187830aeee41f1f8c34ee1c3d669092614373a58be83eb26d32271
Author: Shriya <PES1UG24AM270>
Date:   1776344734

    Add farewell file

commit 1220be45c909e2230e15e7703dbca40093d1763c5d7eb761aa5a206a02593ba3
Author: Shriya <PES1UG24AM270>
Date:   1776344714

    Add world to hello.txtiple files

commit 901aaa23738b1e0502d05759b84c18a08a162d3003fbdcaf154789d54d519f7e
Author: Shriya <PES1UG24AM270>
Date:   1776344698

    Initial commit: Add hello.txtd hello.txtmmitA
```

4B

```
shriya@DESKTOP-6VL9DB0:/tmp/pes_test_phase4$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/0b/15a9dcaa89cbdbbd7a537892e93ce966ef17327d0cee29682c53a6fad25f50
.pes/objects/12/20be45c909e2230e15e7703dbca40093d1763c5d7eb761aa5a206a02593ba3
.pes/objects/17/70f0e189d1320f83cf2581df48e92c3d9234cbf32ae9a63193ff2c43be6006
.pes/objects/1d/275db007bdd4e99d7f4ad1289a65278aec5f582048d6332bed5f2cf7fede70
.pes/objects/33/8f6f809ca34850937699d117ce208f1a688b9a2892702cd3f55dd2a7dae142
.pes/objects/42/abc232674866ceb243e0397955b4d43112de0dc86ca19e2776d9d0599697ca
.pes/objects/57/6426c0f5bb20d6afcb2243f286b17d9aea96fad25b082e430e21e743efb607
.pes/objects/62/250d2d31fb60626bcf6fbe7f2292c330323fb113a20bb4c9d34a10ef5d20ca
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/69/30111d68dd98e7d3e77fbcfe39120d7b1656f7ab5ed3b54e52c4a76d9aa74b
.pes/objects/6a/b21510793a4c4b3dcb39401e79f7bb0dca3c74d1f42fc647ca296ded7b38df
.pes/objects/90/1aaa23738b1e0502d05759b84c18a08a162d3003fbdcaf154789d54d519f7e
.pes/objects/9c/44176122e919122b1fbdcef57aff338580c6cb56facf93d2a1602e4455c794
.pes/objects/cf/52ffc93e187830aeee41f1f8c34ee1c3d669092614373a58be83eb26d32271
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/refs/heads/main
```

4C

```
shriya@DESKTOP-6VL9DB0:/tmp/pes_test_phase4$ echo "=== HEAD ==="
=== HEAD ===
shriya@DESKTOP-6VL9DB0:/tmp/pes_test_phase4$ cat .pes/HEAD
ref: refs/heads/main
shriya@DESKTOP-6VL9DB0:/tmp/pes_test_phase4$ echo -e "\n=== Main branch ==="

=== Main branch ===
shriya@DESKTOP-6VL9DB0:/tmp/pes_test_phase4$ cat .pes/refs/heads/main
576426c0f5bb20d6afcb2243f286b17d9aea96fad25b082e430e21e743efb607
```

Final- Full Integration (make test-integration)

```
shriya@DESKTOP-6VL9DB0:/mnt/c/Users/Shriya/Desktop/PES1UG24AM270-pes-vcs$ make test-integration
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
commit.c:316:12: warning: 'validate_commit_message' defined but not used [-Wunused-function]
  316 | static int validate_commit_message(const char *message) {
      |            ^~~~~~~~~~~~~~~~~~~~~~~
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)
```

```
--- First Commit ---
DEBUG: Starting commit_create
DEBUG: Loaded index with 2 entries
DEBUG: Calling tree_from_index...
DEBUG: Tree hash: 7ac26b11e1ebc08c1c64cbb8c88d2b6845818df5911b9414ae442c0a0c3932f9
DEBUG: No parent (first commit)
DEBUG: Author: Shriya <PES1UG24AM270>
DEBUG: Timestamp: 1776361868
DEBUG: Message: Initial commit
DEBUG: Calling commit_serialize...
DEBUG: Serialized data length: 170 bytes
DEBUG: Serialized content:
tree 7ac26b11e1ebc08c1c64cbb8c88d2b6845818df5911b9414ae442c0a0c3932f9
author Shriya <PES1UG24AM270> 1776361868
committer Shriya <PES1UG24AM270> 1776361868

Initial commit
DEBUG: Calling object_write for commit...
DEBUG: Commit hash: d71a338f7cd62bcc4ba5eb3b1b8d84ac4239dff087e62df2fb0c84758fe54fe0
DEBUG: Calling head_update...
[main d71a338f7cd62bcc4ba5eb3b1b8d84ac4239dff087e62df2fb0c84758fe54fe0] Initial commit
Committed: d71a338f7cd6... Initial commit

Log after first commit:
commit d71a338f7cd62bcc4ba5eb3b1b8d84ac4239dff087e62df2fb0c84758fe54fe0
Author: Shriya <PES1UG24AM270>
Date:   1776361868

    Initial commit
```

```
--- Second Commit ---
DEBUG: Starting commit_create
DEBUG: Loaded index with 2 entries
DEBUG: Calling tree_from_index...
DEBUG: Tree hash: 9619f812fff9f4d450fad5a997d4659f319ca7e89746c8f55f6ad2141a434d94
DEBUG: Parent hash: d71a338f7cd62bcc4ba5eb3b1b8d84ac4239dff087e62df2fb0c84758fe54fe0
DEBUG: Author: Shriya <PES1UG24AM270>
DEBUG: Timestamp: 1776361868
DEBUG: Message: Update file.txt
DEBUG: Calling commit_serialize...
DEBUG: Serialized data length: 243 bytes
DEBUG: Serialized content:
tree 9619f812fff9f4d450fad5a997d4659f319ca7e89746c8f55f6ad2141a434d94
parent d71a338f7cd62bcc4ba5eb3b1b8d84ac4239dff087e62df2fb0c84758fe54fe0
author Shriya <PES1UG24AM270> 1776361868
committer Shriya <PES1UG24AM270> 1776361868

Update file.txt
DEBUG: Calling object_write for commit...
DEBUG: Commit hash: 01cda5ce160d83d3c3b18fcdec30f50f33e925d3ed859ae18bbaa13b739639b8
DEBUG: Calling head_update...
[main 01cda5ce160d83d3c3b18fcdec30f50f33e925d3ed859ae18bbaa13b739639b8] Update file.txt
Committed: 01cda5ce160d... Update file.txt

--- Third Commit ---
DEBUG: Starting commit_create
DEBUG: Loaded index with 3 entries
DEBUG: Calling tree_from_index...
DEBUG: Tree hash: e963f3df809c89eb0e6af46ed6756e9bf7510c2d744b751173d6f414f709bccb
DEBUG: Parent hash: 01cda5ce160d83d3c3b18fcdec30f50f33e925d3ed859ae18bbaa13b739639b8
DEBUG: Author: Shriya <PES1UG24AM270>
DEBUG: Timestamp: 1776361868
DEBUG: Message: Add farewell
DEBUG: Calling commit_serialize...
DEBUG: Serialized data length: 240 bytes
DEBUG: Serialized content:
tree e963f3df809c89eb0e6af46ed6756e9bf7510c2d744b751173d6f414f709bccb
parent 01cda5ce160d83d3c3b18fcdec30f50f33e925d3ed859ae18bbaa13b739639b8
author Shriya <PES1UG24AM270> 1776361868
committer Shriya <PES1UG24AM270> 1776361868
```

```
Add farewell
DEBUG: Calling object_write for commit...
DEBUG: Commit hash: 00490d46931350562217a6312cb8bb0127c6741d7a27b93aa20c73982272be88
DEBUG: Calling head_update...
[main 00490d46931350562217a6312cb8bb0127c6741d7a27b93aa20c73982272be88] Add farewell
Committed: 00490d469313... Add farewell

--- Full History ---
commit 00490d46931350562217a6312cb8bb0127c6741d7a27b93aa20c73982272be88
Author: Shriya <PES1UG24AM270>
Date:   1776361868

    Add farewell

commit 01cda5ce160d83d3c3b18fcdec30f50f33e925d3ed859ae18bbaa13b739639b8
Author: Shriya <PES1UG24AM270>
Date:   1776361868

    Update file.txt

commit d71a338f7cd62bcc4ba5eb3b1b8d84ac4239dff087e62df2fb0c84758fe54fe0
Author: Shriya <PES1UG24AM270>
Date:   1776361868

    Initial committial commitES1Q


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
00490d46931350562217a6312cb8bb0127c6741d7a27b93aa20c73982272be88
```

```
--- Object Store ---
Objects created:
10
.pes/objects/00/490d46931350562217a6312cb8bb0127c6741d7a27b93aa20c73982272be88
.pes/objects/01/cda5ce160d83d3c3b18fcdec30f50f33e925d3ed859ae18bbaa13b739639b8
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/7a/c26b11e1ebc08c1c64cbb8c88d2b6845818df5911b9414ae442c0a0c3932f9
.pes/objects/96/19f812fff9f4d450fad5a997d4659f319ca7e89746c8f55f6ad2141a434d94
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d7/1a338f7cd62bcc4ba5eb3b1b8d84ac4239dff087e62df2fb0c84758fe54fe0
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
.pes/objects/e9/63f3df809c89eb0e6af46ed6756e9bf7510c2d744b751173d6f414f709bccb

=== All integration tests completed ===
```

PHASE 5 AND 6

Q5.1: Implementing pes checkout <branch>

Files that need to change in .pes/:

- HEAD gets rewritten to contain ref: refs/heads/<branch> instead of the current reference
- .pes/index gets rebuilt to reflect the target branch's tree

What must happen to the working directory: you read the target branch's commit hash from .pes/refs/heads/<branch>, then read that commit object to get its tree hash, then recursively walk the tree object to get every blob and its path. For each file, you write the blob's contents to the working directory at that path, creating directories as needed. Files that exist in the current branch but not the target branch must be deleted.

What makes it complex: you have to handle three categories of files simultaneously — files only in the current branch (delete them), files only in the target branch (create them), and files in both branches (overwrite only if content differs). You also need to handle directory creation and deletion, and crucially, you must check for conflicts before touching anything, not halfway through, otherwise you leave the working directory in a broken half-switched state.

Q5.2: Detecting dirty working directory conflicts

The algorithm uses three sources: the index, the object store, and the working directory stat data.

For each file tracked in the index, compare mtime and size against the actual file on disk — this is the same fast-diff that index_status already does. If either differs, the file is locally modified. Next, look up the

same path in the target branch's tree. If the blob hash in the target tree differs from the blob hash in the index for that file, then switching branches would overwrite the user's local change. That's the conflict — refuse checkout and report which files are affected.

You never need to re-hash file contents. The mtime/size check detects local modifications cheaply, and comparing object hashes between the current index and target tree detects whether the branch switch would actually affect that file.

Q5.3: Detached HEAD and recovery

In detached HEAD state, HEAD contains a raw commit hash like a1b2c3d4... instead of ref: refs/heads/main. When you make new commits, each commit is written to the object store correctly and chains to its parent — but no branch ref is updated to point at them. The commits exist but are unreachable from any named branch.

If you switch to another branch without saving your work, those commits become unreferenced and are invisible to normal commands. Recovery is possible because the objects are still in the object store until garbage collection runs. Git maintains a reflog (a log of every position HEAD has pointed to) which lets you find the lost commit hash. In PES-VCS without a reflog, you'd have to scan every object file in .pes/objects/, parse each one that's a commit type, and look for commits whose parent chain doesn't connect to any known branch — then create a new branch pointing at the tip you want to recover.

Q6.1: Garbage collection algorithm

This is a mark-and-sweep approach:

Mark phase: Start from every branch ref in .pes/refs/heads/ and from HEAD. For each ref, read the commit object it points to, add its hash to a "reachable" set. Then follow its parent hash to the previous commit, and its tree hash to the tree object. Recursively walk every tree, adding all subtree and blob hashes to the reachable set. Continue up the entire commit chain until you reach the root commit with no parent.

Sweep phase: List every object file under .pes/objects/. For each one, reconstruct its hash from its path. If that hash is not in the reachable set, delete the file.

The best data structure for the reachable set is a hash set (like a C unordered_set or a simple open-addressing hash table) keyed on the 32-byte ObjectID, giving O(1) insert and lookup.

For 100,000 commits across 50 branches: each commit references one tree, and each tree might reference ~10 blobs on average plus subtrees. A rough estimate is 100,000 commits + 100,000 trees + ~500,000 blobs = roughly 600,000 objects to visit in the worst case. In practice far fewer because blobs are deduplicated across commits when files don't change.

Q6.2: GC race condition with concurrent commit

Here's the exact race:

1. A commit operation runs object_write for a new blob — the blob is written to .pes/objects/ab/cd...

2. GC starts its mark phase, walks all reachable objects from current branch tips. The new blob isn't reachable yet because the commit object referencing it hasn't been written yet
3. GC's sweep phase runs and deletes the blob since it appears unreachable
4. The commit operation now writes the commit object referencing that blob hash — but the blob is gone. The repository is now corrupt

Git avoids this by using a grace period — it never deletes any object newer than a threshold (typically 2 weeks by default). Since a concurrent commit takes milliseconds, a newly written object will always be younger than the grace period and GC skips it. Git also writes a lockfile before GC begins, and commit operations check for this lock. Additionally, Git's pack-refs and gc operations are designed to be re-runnable — if something goes wrong mid-GC, the next run cleans up correctly rather than leaving the repo corrupt.